

T190 derivation depth attempt v0.1 — structural reading:
 $m_\mu/m_e = (24/\pi^2)^{C_2} = (N_c \cdot |W(B_2)| / \pi^2)^{C_2}$ where
 $24 = N_c \cdot 8 = \text{color} \times \text{Weyl-group-order of } B_2$. The cleaner
 substrate-natural form than $24 = \text{rank}^3 \cdot N_c$. Multi-week explicit
 Bergman kernel matrix element computation deferred to Elie;
 this v0.1 makes the structural ingredients explicit.

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2026-05-30 Saturday 11:45 EDT (date-verified)

T190 derivation depth attempt v0.1

0. The target

T190: $m_\mu/m_e = (24/\pi^2)^6 \approx 206.7612$ (vs observed 206.7683, precision 0.0034%).

Goal: structural derivation of this closed form from substrate Bergman kernel integrals on the lepton sector.

1. Cleaner structural reading: $24 = N_c \cdot |W(B_2)|$

The 24 in T190 has multiple substrate-natural decompositions:

Decomposition	Structural meaning
$24 = \text{rank}^3 \cdot N_c = 8 \cdot 3$	arithmetic product (my initial reading)
$24 = 4! =$	S_4
$24 = C_2 \cdot (C_2 - 2) = 6 \cdot 4$	Casimir product (no clear mechanism)
$**24 = N_c \cdot$	$W(B_2)$

The last form is the CLEANEST substrate-natural reading: - $N_c = 3 = \text{color count} / \text{dual Coxeter}$. - $|W(B_2)| = 8 = \text{order of the Weyl group of } B_2 = 2^2 \cdot 2!$ (Weyl group of B_n has order $2^n \cdot n!$). - Product = number of Weyl-orbit positions weighted by color.

So T190 reads:

$$m_\mu/m_e = (N_c \cdot |W(B_2)| / \pi^2)^{C_2}$$

where: - $N_c \cdot |W(B_2)|$ is the “substrate-natural rate base”: color $\times$ Weyl-orbit count. - π^2 is the normalization from 2 angular integrations in the Bergman/Hardy kernel. - C_2 is the gauge-mediator adjoint Casimir (Resolution B per v0.2).

2. Why $N_c \cdot |W(B_2)|$ structurally?

The Weyl group $W(B_2) = D_4$ (dihedral group of order 8) acts on the root system of B_2 by reflections + rotations. It generates all symmetries of the root system / weight lattice.

Per Weyl character formula: for a rep V_λ of a Lie group G , the character is

$$\chi_\lambda = (1/|W|) \cdot \sum_{w \in W} \text{sign}(w) \cdot (\text{Weyl-weight-sum})$$

The $1/|W|$ normalization appears in dimension formulas via the Weyl dimension formula. So $|W(B_2)| = 8$ is the “Weyl-orbit normalization” that appears in K-type character computations.

For substrate matrix elements: a K-equivariant matrix element $\langle V_\lambda | O | V_\mu \rangle$ involves an integration over the K-orbit, weighted by the Weyl group. The factor $|W(B_2)| = 8$ naturally appears.

Combined with $N_c = 3 = \text{color/dual-Coxeter}$ (the substrate’s central charge of the gauge sector), $N_c \cdot |W(B_2)| = 24$ is the substrate-natural matrix-element scaling per radial level.

3. Why π^2 ?

The substrate’s Bergman / Hardy kernel involves π powers from angular integrations on the Shilov boundary $S = (S^4 \times S^1)/Z_2$:

- Volume of $S^4 = 8\pi^2/3$.
- Volume of $S^1 = 2\pi$.
- Volume of $S^4 \times S^1 = 8\pi^2 \cdot 2\pi / 3 = 16\pi^3/3$.
- Volume of $S = 8\pi^3/3$ (after Z_2 quotient).

The π^2 in the denominator of T190 could come from 2 specific angular integration steps in the Bergman kernel matrix element computation. Multi-week explicit verification.

4. Why exponent $C_2 = 6$?

Per Resolution B (Quasi-Eigentone v0.2): the exponent $C_2 = 6$ is the GAUGE-MEDIATOR adjoint Casimir. For the $e \rightarrow \mu$ transition mediated by the gauge sector (W boson, or substrate-equivalent), the gauge mediator’s Casimir enters as the exponent in the closed-form ratio.

Structural mechanism: a single substrate-natural kernel integral per generation step gives the BASE $(24/\pi^2)$; applying it $C_2 = 6$ times (once per “gauge mediator quantum”) gives the final ratio $(24/\pi^2)^6$.

Or: the $C_2 = 6$ emerges from the lepton mass operator’s eigenvalue structure on the spinor radial tower: - Spinor radial tower has Casimirs $5/2, 21/2, 45/2, \dots$ - The mass operator’s matrix element between consecutive levels might pick up a factor $(24/\pi^2)$ per RADIAL step. - $C_2 = 6 = \text{number of “radial-step” applications between } e \text{ and } \mu \text{ generations.}$

For $C_2 = 6$: this could be the number of “kernel propagator steps” in the $e \rightarrow \mu$ transition’s substrate path. Multi-week explicit derivation.

5. The conjectured derivation chain (multi-week)

If the structural reading is correct, the explicit derivation chain is:

1. Set up the substrate Bergman kernel matrix element $\langle V_{(1/2,1/2)} | M | V_{(1/2,1/2)} \rangle$ for the lepton mass operator M .
2. Show the matrix element per radial level evaluates to $(N_c \cdot |W(B_2)| / \pi^2) = 24/\pi^2$.
3. Show the $e \rightarrow \mu$ transition involves $C_2 = 6$ radial-tower / gauge-mediator steps.
4. Result: $m_\mu/m_e = (24/\pi^2)^6$.

Each step is multi-week explicit Bergman kernel computation. The KEY STRUCTURAL CLAIM (this v0.1): the closed form $(24/\pi^2)^{C_2}$ is naturally substrate-derived via $(N_c \times |W(B_2)| / \pi^2)$ per step, C_2 steps total.

6. Cross-check with T2003

T2003 $m_\tau/m_e = 49 \cdot 71 = 3479$. If the same mechanism applied uniformly: m_τ/m_e should be $(24/\pi^2)^k$ for some k . $-(24/\pi^2)^k = 3479 \rightarrow k = \log(3479)/\log(2.43) \approx 9.18$. Not clean integer.

So T2003 has DIFFERENT structural form than T190 — the $e \rightarrow \tau$ transition uses a different substrate-natural mechanism: $-49 = g^2 = 7^2$ (substrate-natural, signature squared). $-71 = ?$ (currently unidentified substrate primary; could be a Mersenne residue or specific prime).

Multi-week investigation of T2003: identify 71's substrate-natural origin.

The fact that T190 ($e \rightarrow \mu$) and T2003 ($e \rightarrow \tau$) have different closed-form structures suggests EACH generation transition has its own substrate-natural mechanism, NOT a uniform per-generation formula. This is consistent with Grace's intermediate-Casimir mapping $\{0, 4, 6\}$ for the three channels — different intermediates $\rightarrow$ different closed forms.

7. Connection to v0.3 quasi-eigentone framework

Per v0.3 4-factor decay rate framework: $-\Gamma(\mu \rightarrow e \bar{\nu} \nu) = (\text{Hall overlap})^2 \times \text{phase-space} \times \text{Bergman kernel integral} \times (\text{gauge mediator})^{\{C_2\}}$. - T190's closed form $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$ fits as the KERNEL-INTEGRAL $\times$ GAUGE-MEDIATOR product (with overlap and phase-space contributing to absolute rate).

8. Honest scope + tier

RIGOROUS (existing math): - T190 closed form $(24/\pi^2)^6$ (Casey/BST, 0.004% precision verified). $-24 = N_c \cdot |W(B_2)|$ (substrate primary $\times$ Weyl group order, arithmetic). - Bergman kernel $c_{FK} = 225/\pi^{(9/2)}$ for D_{IV}^5 (T2442). - Weyl group $W(B_2) = D_4$ of order 8.

STRUCTURAL READING (v0.1): T190 $= (N_c \cdot |W(B_2)| / \pi^2)^{\{C_2\}}$ where the base is color $\times$ Weyl-orbit / π^2 (substrate-natural matrix-element scaling per radial level) and exponent is gauge-mediator adjoint Casimir.

CONJECTURED DERIVATION (multi-week): 4-step chain (setup kernel matrix element $\rightarrow$ evaluate per-step factor as $24/\pi^2 \rightarrow$ show $C_2 = 6$ step count for $e \rightarrow \mu \rightarrow$ assemble closed form). Each step is multi-week.

Cal #27 / honesty: this is a DEPTH ATTEMPT, not a derivation. The structural reading $24 = N_c \cdot |W(B_2)|$ is cleaner than the arithmetic $24 = \text{rank}^3 \cdot N_c$ (both are correct factorizations; the Weyl group reading is more structurally meaningful). The conjectured derivation chain identifies what would need to be computed; actual computation is multi-week. T2003 ($m_\tau/m_e = 49 \cdot 71$) has DIFFERENT structure — each generation transition has its own substrate-natural mechanism.

Routed: $\rightarrow$ Elie: explicit Bergman kernel matrix element computation on spinor radial tower $V_{(k+1/2, 1/2)}$ is the multi-week target; the v0.1 framework predicts the per-step matrix element $= N_c \cdot |W(B_2)| / \pi^2 = 24/\pi^2$. $\rightarrow$ Keeper: depth-shift attempt filed; structural reading $24 = N_c \cdot |W(B_2)|$ identified; explicit derivation multi-week. $\rightarrow$ me: continuing depth work or shifting back to sweep items per Casey direction.

— Lyra, T190 derivation depth attempt v0.1. STRUCTURAL READING: T190 $m_\mu/m_e = (24/\pi^2)^6 = (N_c \cdot |W(B_2)| / \pi^2)^{\{C_2\}}$ where $24 = \text{color} \times \text{Weyl-group-order of } B_2$ (cleaner than arithmetic $24 = \text{rank}^3 \cdot N_c$). Exponent $C_2 = 6$ is gauge-mediator adjoint Casimir (Resolution B). π^2 from 2 angular kernel integrations. Conjectured derivation chain: 4-step explicit Bergman kernel matrix-element computation, multi-week. T2003 ($m_\tau/m_e = 49 \cdot 71$) has different structure (each generation transition has its own substrate-natural mechanism). v0.3 4-factor decay framework provides the scaffold for this derivation.